FAHMAI BENCHMARK · TRILOGY FINALE · 2026

# FahMai The Finale

## Enterprise Data-Agent Showdown

Two fiscal years of a Thai retailer's operations. 100+ questions. One agent.

Stage 1 · Kaggle | 3 Tracks | Multimodal | Thai / EN | Agentic


Launch: Mon · Private LB: Thu 4 Jun 2026, 08:00

# The Trilogy

FahMai has been the spine of the SuperAI benchmark series.

LEVEL 1

## FahMai RAG Hack

Retrieve over the product directory.

- 127 products across 5 in-house brands
- Thai customer questions
- Single-table or KB lookup
- Tests: retrieval + answer extraction

LEVEL 2

## Directory Q&A

Bigger directory + refusal hygiene.

- Larger CSV — doesn't fit in context
- Out-of-corpus refusal (exact phrase)
- Public LB + Private LB split
- Tests: filter / tool-use + clean refusal

LEVEL 3 · FINALE

## Enterprise Agent

Two fiscal years of REAL operations.

- 31 tables · 53K docs · 6.1K renders
- Bitemporal · schema cutover · DQ artifacts
- 100 public + 50 private questions
- 2 tracks (Local / ThaiLLM)
- Tests: cross-modal reasoning at scale

# The FahMai Universe

A Thai multi-channel consumer-electronics retailer. 2 fiscal years (2024–2025).

### THE COMPANY

ฟ้าใหม่ (FahMai) — a multi-channel electronics retailer with physical stores, an online store, and a B2B reseller arm. The data lake holds two fiscal years of operations: every POS line item, every bank statement, every customer chat, every policy memo, every quarter-close report.

### FIVE HOUSE BRANDS

| | |
|---|---|
| **SaiFah** | Phones · Tablets |
| **DaoNuea** | Laptops · Computers |
| **KluenSiang** | Audio |

### OPERATIONS SURFACE

- **10** physical & remote branches
- **13+** vendors & 4 partner brands
- **14** operating bank accounts
- **100K+** customers / 500K+ POS rows

# The Data — Overview

One bundle. Five surfaces. ~4 GB. Everything an analyst at FahMai would actually have.

### TABLES

**31**

DIM + FACT

Dimension & fact CSVs. Covers 2024-01 → 2025-12. Includes mid-2025 schema cutover.

### DOCS

**53K**

narrative

Memos, meeting minutes, emails, knowledge base, LINE OA + LINE Works chats.

### RENDERS

**6.1K**

PNG / PDF

Bank statements, receipts, invoices, warranty forms, lease + training PDFs, promo banners.

### LOGS

**7.9K**

operational

POS .tsv per branch-day, web .jsonl per day, PayWise monthly fee CSVs.

### REPORTS

**32**

cadence

Monthly OPS reports. Quarterly FIN close. Already-aggregated views.

Total bundle ≈ 4.0 GB · ~67K files · zipped & ready

# The Data — Tables

31 CSVs. Dimensions describe the world. Facts describe what happened.

## DIM_*

Dimension tables — who, what, where

DIM_PRODUCT · DIM_CUSTOMER · DIM_VENDOR

DIM_BRANCH · DIM_EMPLOYEE · DIM_BANK_ACCOUNT

DIM_PROMO_CAMPAIGN · DIM_POLICY_VERSION

DIM_VENDOR_CONTRACT_VERSION

DIM_SIGNING_AUTHORITY_LADDER

DIM_PRODUCT_RECALL_HISTORY · DIM_DATE · …

## FACT_*

Fact tables — events that happened on a date

FACT_SALES + FACT_SALES_LINE_ITEM

FACT_RETURN · FACT_REFUND_PAID

FACT_WARRANTY_CLAIM · FACT_CS_INTERACTION

FACT_PROMO_REDEMPTION

FACT_BANK_TRANSACTION · FACT_VENDOR_PAYMENT

FACT_LOYALTY_LEDGER · FACT_SHIPPING

FACT_INVENTORY_MONTHLY_SNAPSHOT · …

# The Data — Narrative

Tables describe what; narrative explains why. 53,428 markdown documents.

| MEMO | MEETING MIN. | EMAIL |
|---|---|---|
| 16 | 26 | 25 |
| internal policy memos | minutes of weekly / monthly meetings | all-staff & departmental emails |

| L1 KB | LINE OA chats | LINE WORKS |
|---|---|---|
| — | 37,441 | 15,802 |
| product specs, policies, FAQ entries | customer service chats (B2C) | internal team threads (between staff) |

Most narrative is in Thai. Some in English. A small fraction is mixed.

# The Data — Renders

6,128 rendered artifacts. Scanned-document realism. PNG + PDF.

**BANK STATEMENT**

monthly per account

**RECEIPT**

POS receipts

**VENDOR INVOICE**

vendor billings

**WARRANTY FORM**

warranty claim forms

**E7 BANNER**

promo banners

**T2 DOC (PDF)**

lease, training, etc.

**T3 DOC**

ad-hoc internal forms

## Why renders matter

Some facts live ONLY in the rendered surface. A bank statement's running balance. A signature on a warranty form. A typo in a vendor invoice. Your agent has to OPEN the file.

# What makes it hard

Five sources of difficulty that compound across tiers.

1

### THAI / EN / MIXED

Thai · English · พ.ศ. dates.

2

### MULTIMODAL

Tables ↔ docs ↔ renders ↔ chats.

3

### BITEMPORAL

Right answer depends on the date at the event.

4

### DATA-QUALITY ARTIFACTS

Real ops data has bugs.

5

### SCHEMA CUTOVER

Same table — two shapes.

# The Questions

100 public + 50 private. Authored in the question's language.

PUBLIC (Kaggle from Day 1)

100 questions

PRIVATE (Thu 4 Jun 08:00)

50 questions

FIVE DIFFICULTY TIERS

from quick lookups → forensic reconciliations

Sizes, labels, and the texture of each are yours to discover.

# Example — Easy

One question. Multiple data surfaces. No shortcut.

easy

THE QUESTION

ในวันที่ 15 ธันวาคม 2024 ลูกค้าสามารถคืนสินค้าได้ภายในกี่วันตามนโยบายของฟ้าใหม่ที่มีผลบังคับใช้ ณ วันนั้นครับ

WHAT IT TESTS

- Date-aware lookup against a policy table
- Read effective_date / end_date lifecycle
- Resolve at the question's as-of date, not the latest row

DATA SOURCES

DIM_POLICY_VERSION
filter by policy_variable = return_window_days
resolve at 2024-12-15
(≈ 1 query · 1 table)

# Example — Medium

One question. Multiple data surfaces. No shortcut.

medium

THE QUESTION

ขอเปรียบเทียบผลแคมเปญ 11.11 Mega Sale ปีต่อปีหน่อยครับ โดยใช้ข้อมูลจาก FACT_PROMO_REDEMPTION เทียบ MEGA-1111-2567 กับ MEGA-1111-2568: (1) จำนวน redemption รวมของปี 2567, (2) ยอดส่วนลดรวมของปี 2567, (3) จำนวน redemption รวมของปี 2568, (4) ยอดส่วนลดรวมของปี 2568

WHAT IT TESTS

- Filter by campaign_id across two years
- Group + aggregate (count + sum)
- Return a multi-part numeric answer (4 numbers)

DATA SOURCES

FACT_PROMO_REDEMPTION
↔ DIM_PROMO_CAMPAIGN (resolve campaign IDs)
group by campaign_id
sum(discount_thb), count(*)

# Example — Hard

One question. Multiple data surfaces. No shortcut.

hard

THE QUESTION

วันที่ 2025-04-05 ใน LINE WORKS thread ของทีม Finance / Ops มีการแจ้งเคส invoice ID ซ้ำหลัง schema cutover ของ vendor V-013 (PayWise) ขอให้ตอบ 3 ข้อนี้ครับ: (1) invoice ID ที่ถูกแจ้งว่าซ้ำคือเลขอะไร, (2) ใน FACT_VENDOR_PAYMENT มี payment record กี่แถวที่ใช้ invoice ID เดียวกันนี้, (3) แต่ละแถวจ่ายเป็นจำนวนเงินเท่าไหร่ (THB) และ posting_date เป็นวันที่เท่าไหร่

WHAT IT TESTS

- Cross-modal: chat thread + FACT table reconciliation
- Read LINE WORKS markdown for the day, find the alert
- Pull matching rows by invoice_id from FACT table
- Multi-part answer (id + count + amounts + dates)

DATA SOURCES

docs/chat_line_works/ on 2025-04-05
find the duplicate-invoice alert
+ FACT_VENDOR_PAYMENT
filter by vendor_id = V-013 + invoice_id
list paid_amount + posting_date per row

# Example — Extremely Hard

One question. Multiple data surfaces. No shortcut.

`extremely_hard`

## THE QUESTION

ทีม Finance สงสัยว่า app-side redemption log ของ SF-LAUNCH-2568 ช่วง 2025-07-15 ถึง 2025-07-31 มี double-logging. ขอให้ report เป็น 5 ตัวเลข: (1) จำนวน redemption ทั้งหมดใน FACT_PROMO_REDEMPTION ภายใต้ campaign นี้, (2) จำนวน phantom redemption (txn_id ซ้ำข้าม channel), (3) จำนวน unique หลังหัก phantom, (4) net discount cost ที่ตรงกับ FACT_SALES.discount_total_thb ของ cohort, (5) net revenue ของ cohort ⟶ ROI กี่เท่า. พร้อม cross-check FACT_BANK_TRANSACTION ของ V-013 ในเดือน 2025-07.

## WHAT IT TESTS

- Bitemporal: campaign window + per-row date resolution
- Detect & dedupe duplicate rows across channels
- Multi-table reconciliation (promo + sales + bank)
- Compute a derived ratio (ROI) over a filtered cohort

## DATA SOURCES

FACT_PROMO_REDEMPTION (campaign + dedup)
FACT_SALES (cohort + discount + revenue)
FACT_BANK_TRANSACTION (V-013, July 2025)
5+ surfaces · 5 reconciliations

# Evaluation — Public Kaggle

100 questions. Submit a CSV. Score on accuracy. Live leaderboard.

## SCORING

### Accuracy (0 to 1)

- Each question graded independently
- Token-based pass/fail — deterministic grader
  must_contain_any_of (AND-of-ORs)
  must_not_contain (forbidden tokens)
- Submit submission.csv → 100 rows
- Column response holds your agent's final answer
- Grade locally: python grade.py submission.csv train_labels.json

## BASELINE

### Codex CLI · gpt-5.5-xhigh

Minimal-prompt agentic loop

**86 / 100**
correct

**$94**
API-equivalent cost

**2.9 hr**
wall-clock @ 3 workers

**78.7 M**
tokens (93% cached)

# Baseline — Token usage per tier

Codex / gpt-5.5-xhigh on the public 100. Median ≠ mean — tails matter.

| tier | median tokens | mean tokens | mean cost (USD) |
|---|---|---|---|
| easy | 98,916 | 159,646 | $ 0.28 |
| medium | 160,427 | 434,330 | $ 0.56 |
| hard | 768,413 | 1,003,169 | $ 1.15 |
| extremely_hard | 1,225,501 | 1,766,424 | $ 1.97 |
| special | 408,765 | 708,379 | $ 0.93 |

$5 / 1M input · $0.50 / 1M cached input · $30 / 1M output (most input cached across a corpus-rereading agent loop)

# Evaluation — Private Leaderboard

The real ranking. Released mid-week. Same scoring. Tougher questions.

PRIVATE-LB RELEASE

**Thursday 4 June 2026 · 08:00 ICT**

### 50 QUESTIONS

**hard · extremely_hard · special***

- Same data bundle — no new files
- Same scoring scheme as public
- Final ranking = private LB only
- No re-submission after deadline

### WHAT TO PREPARE

- Pipeline that reads your data bundle once
- Re-runnable end-to-end in < 4 hours
- Output: submission.csv with 50 rows
- Reproducible from a single command
- Cost telemetry recorded
- Track-specific gateway live by Thu morning

# Scoring — Beat the Baseline

Lower cost than baseline → bonus. Higher cost → penalty. Same formula across all 3 tracks.

ACCURACY + BASELINE-EFFICIENCY BONUS

```
final = accuracy + α · (1 - your_cost / baseline)
```

α = 0.20 · bonus when under baseline, penalty when over · clipped to [0, 1]

SINGLE RANKING

REWARDS EFFICIENCY

CROSS-TRACK

PER-TRACK CURRENCY · BASELINE

API → USD ($94 Codex baseline)

Local → tokens (organizer reference)

ThaiLLM → tokens (organizer reference)

# Two Tracks

Same data. Same questions. Different model constraint. Both run on NTi compute — no playground / API allowed.

TRACK 1

## ThaiLLM-based

Locally-deployed Thai LLMs only. No playground / API.

NTi · 1× B200 · or LANTA

- Models trained primarily in Thai
- Must be hosted locally on the NTi GPU
- No remote-API fallback (OpenAI, Anthropic, ...)
- Showcase the Thai-LLM ecosystem
- Special pitch-prize candidate

TRACK 2

## Open-Weight Local

Any open-weight model that fits on the provided GPU.

NTi · 1× B200 · or LANTA

- Any open-weight model (Llama, Qwen, Mistral, ...)
- Self-hosted inference on the NTi GPU
- Same anti-cheat — no remote-API fallback
- Cost = tokens generated (per the scoring formula)

All Agentic inference runs on NTi compute (1× B200 or LANTA). OCR sub-track uses a separate RTX 5090 · 32 GB VRAM (see next slide).

# OCR Sub-track

Pull structured fields out of bank-statement renders.  Fits on a single 32 GB GPU.  Separate leaderboard.

SCOPE

## Bank-statement sample

RTX 5090 · 32 GB VRAM

- Sampled from the rendered bank-statement set
- Constraint: single 32 GB GPU (RTX 5090)
- Submit a JSONL of extracted fields per image

Optional data for testing / research:
huggingface.co/spaces/teppap/research

METRIC

## Edit-Distance + Structure

```
Levenshtein edit distance
+
Field-level structure accuracy
```

\+ Efficiency:  back-test against bank-statements
from OUTSIDE the public sample
→ tests generalization, not memorization

Fields:  amount · date · account · description

# Back-test — Endpoints

How we score your submission. Anti-cheat enforced. Audit trail captured.

## OCR BACK-TEST

INPUT

Image / PDF · base64-encoded

OUTPUT

JSON (structured fields)

ENDPOINT

`POST http://[NTi_IP]/ocr`

Runs on RTX 5090 · 32 GB VRAM

## AGENTIC BACK-TEST

Local + ThaiLLM tracks

INPUT

```
{ "question": "..." }
```

OUTPUT

```
{ "id": "...",
  "answer": “...”,
  "total_output_token": N }
```

ENDPOINTS

`POST http://[NTi_IP]/agent/local`

`POST http://[NTi_IP]/agent/thaillm`

1× B200 / LANTA · audit-trail captured

โกง = ตัดสิทธิ์ · Cheating disqualifies · Audit trail of LLM thinking captured on every request

# Timeline

Mon kickoff → Thu pitch finals. Six back-test simulation slots Wednesday.

## MON

16:30

แจกโจทย์

20:30

Kaggle opens (OCR + Agentic)

## TUE

13:00

Consult session

## WED

13:00

Back-test sim #1

14:30

Back-test sim #2

16:00

Back-test sim #3

19:00

Back-test sim #4

20:30

Back-test sim #5

22:00

Back-test sim #6

23:59

OCR Kaggle CLOSES

## THU

08:00 – 11:00

Agentic back-test window

09:00

Pitchers announced

12:00

Slide + code on mysuper
i. Presentation Slide (PDF only)
ii. Code (.zip with the following directory: backend, other)
iii. All LLM Audit Trail (.zip with {id}.txt)

13:30

Pitching (10 + 8 min)



Pitching = 10 min present + 8 min Q&A. Final rubric on the next slide.

# Final Pitch — Rubric

Final ranking combines numeric leaderboard standing with a live solution pitch.

LEADERBOARDS · 50% → numeric performance

### Private LB

40 %

50 held-out questions released Thu 4 Jun 08:00 ICT.
Final numeric ranking — hard / extremely_hard / special tiers.

### OCR Private LB

10 %

Sub-track on ~6,000 rendered images + Back test
Separate leaderboard, scored on the private-side split.

PITCH · 50% → live solution presentation

### Solutions (harness + defense)

30 %

- Architecture · retrieval · agentic harness
- Model choice · cost / accuracy tradeoffs
- Adversarial defense — how the agent handles attempts to override its instructions (in the corpus and in the prompt). Concrete examples.
- What surprised you in the data · lessons

### Presentation + demo

20 %

- 10-min slot — present + Q&A
- Live demo encouraged
- Reproducibility shown (1-command run)
- Clarity, narrative, team chemistry

# Kaggle Links

**Agentic Kaggle:**
https://www.kaggle.com/t/22741abf563441268dd595a03479a954